Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/03/2006
PI Name:	Fritzsch, Bernd Ph.D.		
Project Title:	Neurobiology Of Vestibular Development In Mutant Mice		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
PI Email:	Fritzsch@Creighton.edu	**Fax:**	FY 402-280-5556
PI Organization Type:	UNIVERSITY	**Phone:**	402-280-2915
Organization Name:	Creighton University		
PI Address 1:	2500 California Plaza		
PI Address 2:			
PI Web Page:	http://www2.creighton.edu/		
City:	Omaha	**State:**	NE
Zip Code:	68178	**Congressional District:**	2
Comments:	NOTE (June 2016): Dr. Fritzsch is currently at University of Iowa		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	de Caprona, Dominique (Creighton University) Beisel, Kirk (Creighton University) Nichols, Dave (Creighton University)		
Grant/Contract No.:	NAG2-1595		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To investigate the effect of normal and hypergravity on vestibular development in mice by investigating the effect of absence of otoconia on gravistatic behavior, neuronal connections and sensory epithelia gene expression. Using otoconia mutant mice we can investigate the absence of linear acceleration perception such as gravity on the behavior and connectional development of mice raised in 1g. Such genetically controlled manipulations allow for a more standardized analysis of a greater number of animals. By understanding the effect of absence of linear acceleration perception (gravity) on the functional development of ear input into the behavior and neuronal organization, we will be able to provide a more molecular understanding of the anticipated changes in the absence of gravity.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our results from a study of embryonic rats developing in microgravity as well as several studies using hypergravity exposure suggest that exposure to altered gravity alters the development of gravistatic vestibular connections and causes long lasting behavioral changes. We now propose to examine how vestibular connections are affected by hypergravity and whether there is a critical age during which gravistatic vestibular connections must be exposed to 1g in order to develop normal connections. The connections of specific inner ear receptors to their central nuclei will be analyzed using the lipophilic dye, DiI, to trace fibers in aldehyde fixed tissue and to analyze their connections to vestibular neurons both light and electron microscopically. In a first, descriptive approach, we will examine the normal sequence of development and how it is altered by exposure to hypergravity by exposing developing rat embryos and pups to hypergravity (2g) for variable periods of time before and after birth. Subsequently we will seek a possible critical age during which the formation of abnormal connections between the ear and brain caused by hypergravity can be appropriately reorganized by intermittent exposure to 1g. For this we will expose rat pups to hypergravity with intermittent exposures to 1g to counteract the hypergravity-induced effects. It is hoped that this program will ultimately help to identify the ages during which biologically relevant gravity stimuli are essential for normal vestibular development. Thus it may be possible to minimize long-term effects of microgravity on the development of vestibular connections during long-term flights by using a 1g centrifuge at appropriate intervals.
Task Progress:	Progress on Aim1: Behavioral analysis of vestibular mutant mice. We have finalized data collection and have a paper in press on our behavioral data. This paper shows that mice without otoconia can not only learn behavioral balance tasks but do they with a steeper learning curve than their control littermates owing to the more profound handicap they are starting with. This paper refutes previous claims in the literature that removal of the ear eliminates spatial learning, likely because the sensory projection of the tlt-mutant mice is at least grossly normal whereas it is absent in animals that underwent ear removal. We have now extended this work to ask whether animals are able to move a 30o inclined bar up and down, both in normal light and in the dark. Surprisingly, the tlt mice with no otoconia performed extremely well in that behavioral task. We are currently finalizing these experiments in the no-cost extension period of this grant. If the first impression is supported by our statistics, we have to conclude that linear acceleration perception is for mice not relevant for simple balance tasks in the horizontal and inclined levels, a truly surprising finding. Title: Partial behavioral compensation is revealed in balance tasked mutant mice lacking otoconia. Authors: D. Crapon de Caprona, K.W. Beisel, D.H. Nichols & B.Fritzsch. Brain Research Bulletin, 64 (2004) 289-301. Progress in Aim 2: Development of connections of vestibular end organs to the brainstem and cerebellum. We have shown that the overall projection of tlt mutant mice is at least grossly normal (see the above cited paper) and are currently investigating the details of the saccular projections in neonates compared to the posterior crista. We concentrate on three areas, the projection to the cell group Y (which appears to be even more prominent in tlt mutant mice), the projection to the dorsal part of the medial vestibular nucleus (which appears to be reduced in tlt mutant mice) and the projection the uvula in the cerebellum (which appears to be less segregated than in control mice). We are currently analyzing more of these animals with longer diffusion times. To achieve better segregation of fiber staining we developed in collaboration with PTI Research modified lipophilic dyes. These new techniques have recently been published. *Fritzsch, B., Muirhead, K.A., Feng, F., Gray, B.D., (2005) Ohlsson-Wilhelm, B.M. (2005) Diffusion and imaging properties of three new lipophilic tracers, NeuroVue ™ Maroon, NeuroVue ™ Red and NeuroVue ™ Green and their use for double and triple labeling of neuronal profile. Brain Res. Bull., 66:249-258. Progress in Aim 3: Alterations in ion channel expression in vestibular organs of mutants. We have generated, in collaboration with Dr. B. Soarez, Iowa City, a microchip that allows easy analysis of differences in expression profiles of a large number of ion channels. We also generated probes for a number of ion channels that can now be used for quantitative assessment of ion channel expression (real time PCR) as well as qualitative assessment (in situ hybridization of utricular sensory epithelia). We have therefore now generated all the necessary tools for a high throughput analysis of vestibular endorgans of tilt and otogelin mutant mice and control littermates in the next two years. We have collected several tilt and wildtype mouse ears and are preparing them for microarray analysis.
Bibliography Type:	Description: (Last Updated: 03/26/2008)
Articles in Peer-reviewed Journals	de Caprona MD, Beisel KW, Nichols DH, Fritzsch B. "Partial behavioral compensation is revealed in balance tasked mutant mice lacking otoconia. " Brain Res Bull. 2004 Dec 15;64(4):289-301. PMID: 15561463 , Dec-2004
Articles in Peer-reviewed Journals	Fritzsch B, Muirhead KA, Feng F, Gray BD, Ohlsson-Wilhelm BM. "Diffusion and imaging properties of three new lipophilic tracers, NeuroVue ™ Maroon, NeuroVue ™ Red and NeuroVue ™ Green and their use for double and triple labeling of neuronal profile." Brain Res Bull. 2005 Aug 15;66(3):249-58. PMID: 16023922 , Aug-2005
Articles in Peer-reviewed Journals	Fritzsch B, Pauley S, Matei V, Katz DM, Xiang M, Tessarollo L. "Mutant mice reveal the molecular and cellular basis for specific sensory connections to inner ear epithelia and primary nuclei of the brain." Hear Res. 2005 Aug;206(1-2):52-63. Review. PMID: 16080998 , Aug-2005
Articles in Peer-reviewed Journals	Fritzsch B, Gregory D, Rosa-Molinar E. "The development of the hindbrain afferent projections in the axolotl: evidence for timing as a specific mechanism of afferent fiber sorting." Zoology (Jena). 2005;108(4):297-306. PMID: 16351978 , Oct-2005
Articles in Peer-reviewed Journals	Beisel KW, Wang-Lundberg Y, Maklad A, Fritzsch B. "Development of the mammalian vestibular neurosensory ear." J Vestib Res. 2005, in press , Oct-2005
Books/Book Chapters	Fritzsch B. "Ear and lateral line of vertebrates: organization and development." in "Encyclopedia of Life Sciences" Chichester: John Wiley & Sons, 2005., Oct-2005